

Lab 1: Python Basics for Artificial Intelligence

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1 Objective

To help students understand the basics of Python programming with a focus on concepts that are essential for Artificial Intelligence (AI).

2 Part 1: Simple Python Programs for AI Basics

2.1 Activity 1: Display numbers on screen using Python IDLE

```
1 # Display numbers
2 print(1)
3 print(123)
4 print(3.14)
```

2.2 Activity 2: Display strings on screen

```
1 # Display strings
2 print("Hello, AI students!")
3 print("Python is fun!")
```

2.3 Activity 3: Use Python as a calculator

```
1 # Python as a calculator
2 print("Sum:", 5 + 7)
3 print("Difference:", 10 - 3)
4 print("Product:", 4 * 6)
5 print("Quotient:", 12 / 3)
```

2.4 Activity 4: Integer division and remainder

```
1 # Integer division and remainder
2 a = 17
3 b = 4
4 print("Integer division:", a // b)
5 print("Remainder:", a % b)
```

2.5 Activity 5: Calculate powers

```
1 # Calculate powers
2 print("4^3 =", 4**3)
3 print("4^10 =", 4**10)
4 print("4^29 =", 4**29)
5 print("4^150 =", 4**150)
6 print("4^1000 =", 4**1000)
```

2.6 Activity 6: Operator precedence expressions

```
1 # Math expressions
2 expr1 = 2 + 3 * 4
3 expr2 = (2 + 3) * 4
4 expr3 = 10 / 2 + 6
5 expr4 = 2**3 + 5*2
6
7 print(expr1, expr2, expr3, expr4)
```

2.7 Activity 7: Combine numbers and text

```
1 # Combine numbers and text
2 age = 20
3 print("I am " + str(age) + " years old")
```

2.8 Activity 8: Take input from the keyboard

```
1 # Take input from user
2 name = input("Enter your name: ")
3 print("Welcome, " + name + "!")
```

2.9 Activity 9: Check if a number is even (input from user)

```
1 # Check if number is even
2 num = int(input("Enter an integer: "))
3 if num % 2 == 0:
4     print(num, "is even")
```

2.10 Activity 10: Check if a number is even or odd (input from user)

```
1 # Check if number is even or odd
2 num = int(input("Enter an integer: "))
3 if num % 2 == 0:
4     print(num, "is even")
5 else:
6     print(num, "is odd")
```

2.11 Activity 11: Sum of all values from 0-10 using while loop

```
1 # Sum using while loop
2 i = 0
3 total = 0
4 while i <= 10:
5     total += i
6     i += 1
7 print("Sum of numbers from 0 to 10:", total)
```

2.12 Additional Exercises from previous Lab (Optional)

2.12.1 Hello World Program

```
1 print("Hello, World!")
```

2.12.2 Basic Input/Output

```
1 name = input("Enter your name: ")
2 print("Hello, " + name + "!")
```

2.12.3 Arithmetic Operations

```
1 num1 = float(input("Enter first number: "))
2 num2 = float(input("Enter second number: "))
3 print("Sum:", num1 + num2)
4 print("Difference:", num1 - num2)
5 print("Product:", num1 * num2)
6 print("Quotient:", num1 / num2)
```

2.12.4 Odd or Even Checker

```
1 num = int(input("Enter a number: "))
2 if num % 2 == 0:
3     print(num, "is even")
4 else:
5     print(num, "is odd")
```

2.12.5 List of Squares

```
1 squares = [i**2 for i in range(1,11)]
2 print("Squares:", squares)
```

2.12.6 List Operations

```
1 numbers = [1,2,3,4,5]
2 numbers.append(6)
3 numbers.remove(3)
4 numbers.sort()
5 print("Updated list:", numbers)
```

2.12.7 Dictionary Operations

```
1 students = {"Ali":85, "Sara":92, "Ahmed":78}
2 students["Zara"] = 88
3 students["Ali"] = 90
4 del students["Ahmed"]
5 print("Updated students dictionary:", students)
```

2.12.8 Function to Calculate Area of Circle

```
1 import math
2 def area_of_circle(radius):
3     return math.pi * radius**2
4 r = float(input("Enter radius: "))
5 print("Area of circle:", area_of_circle(r))
```

2.12.9 Lambda Function

```
1 add = lambda x, y: x + y
2 print("Sum using lambda:", add(5,3))
```

2.12.10 List Comprehension for Even Numbers

```
1 even_numbers = [x for x in range(1,21) if x % 2 == 0]
2 print("Even numbers:", even_numbers)
```

3 Part 1: Simple Python Programs for AI Basics

3.1 Hello World Program

```
1 # Hello World Program
2 print("Hello, World!")
```

3.2 Basic Input/Output

```
1 # Take user's name and greet
2 name = input("Enter your name: ")
3 print("Hello, " + name + "!")
```

3.3 Arithmetic Operations

```
1 # Arithmetic operations
2 num1 = float(input("Enter first number: "))
3 num2 = float(input("Enter second number: "))
4
5 print("Sum:", num1 + num2)
6 print("Difference:", num1 - num2)
7 print("Product:", num1 * num2)
8 print("Quotient:", num1 / num2)
```

3.4 Odd or Even Checker

```
1 # Check if number is odd or even
2 num = int(input("Enter a number: "))
3 if num % 2 == 0:
4     print(num, "is even")
5 else:
6     print(num, "is odd")
```

3.5 List of Squares

```
1 # List of squares from 1 to 10
2 squares = []
3 for i in range(1, 11):
4     squares.append(i**2)
5 print("Squares:", squares)
```

3.6 List Operations

```
1 # List operations
2 numbers = [1, 2, 3, 4, 5]
3
4 # Add an element
5 numbers.append(6)
6
7 # Remove an element
8 numbers.remove(3)
9
10 # Sort the list
11 numbers.sort()
12
13 print("Updated list:", numbers)
```

3.7 Dictionary Operations

```
1 # Dictionary of students and marks
2 students = {"Ali": 85, "Sara": 92, "Ahmed": 78}
3
4 # Add a new student
5 students["Zara"] = 88
6
7 # Update marks
8 students["Ali"] = 90
9
10 # Delete a student
11 del students["Ahmed"]
12
13 print("Updated students dictionary:", students)
```

3.8 Function to Calculate Area

```
1 import math
2
```

```

3 # Function to calculate area of circle
4 def area_of_circle(radius):
5     return math.pi * radius**2
6
7 r = float(input("Enter radius: "))
8 print("Area of circle:", area_of_circle(r))

```

3.9 Lambda Function

```

1 # Lambda function to add two numbers
2 add = lambda x, y: x + y
3 print("Sum using lambda:", add(5, 3))

```

3.10 List Comprehension

```

1 # List of even numbers from 1 to 20 using list comprehension
2 even_numbers = [x for x in range(1, 21) if x % 2 == 0]
3 print("Even numbers:", even_numbers)

```